

Forward-Looking, Open Multi-Hazard Flood Screening as Strategic Decision Support for Critical Infrastructure in Under-Resourced Southeast Asian Cities

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Abstract—The Southeast Asian cities most exposed to flooding are precisely those least able to obtain usable hazard information, and they are also where the region’s critical port infrastructure and trade chokepoints concentrate. The only comparable-resolution (30 m) three-hazard regional model is commercial and closed, and its free tranche excludes the major ASEAN megacities; bespoke engineering studies are not reproducible; and open global products are an order of magnitude coarser and omit the pluvial hazard that dominates urban flooding here. We present an open-source, open-data pipeline producing design-event coastal, fluvial and pluvial flood-depth maps at 30 m for Singapore, Kuala Lumpur, Bangkok and Jakarta, under four climate combinations (SSP2-4.5/SSP5-8.5 $\times$ 2050/2100). Every input is freely accessible, and the pluvial hazard is anchored to each national meteorological service’s published Intensity–Duration–Frequency (IDF) standard, closing a 28–62% deficit that global synthetic rainfall carries over tropical-convective extremes. We make the model *trustworthy*, not merely reproducible. A *pre-defined* model-blind hotspot location-skill gate yields statistically significant discriminative skill (true-skill-statistic confidence interval excluding zero) in all four cities; one of four meets the stricter joint sensitivity/specificity PASS bar. A bathtub coastal over-prediction of 1.7–25 $\times$ at RP100 is structurally corrected by a local-inertia shallow-water solver (Bangkok 12.5 $\times$ to $\approx$ 1 $\times$; 3,546 $\rightarrow$ 283 km²). Framed through the lens of climate-adjusted strategic decision-making, the release is offered as a public good for the municipal disaster-management, adaptation and port-planning agencies that cannot license commercial models.

Index Terms—humanitarian technology, flood risk, open data, multi-hazard, decision support, climate adaptation, ports, Southeast Asia.

I. INTRODUCTION

Southeast Asia contains six of the ten cities globally most exposed to coastal flooding by 2100 under high-emission scenarios. Roughly 750 million people live in the ASEAN region, with about a fifth of regional GDP on flood-exposed land [1]. The stressors compound: AR6 median sea-level rise (SLR) under SSP5-8.5 by 2100 ranges from +0.62 m (Singapore) to +1.62 m (Bangkok) [2]; tropical convective rainfall intensifies at or above the Clausius–Clapeyron rate [3]; and several megacities subside at 1–25 cm yr^{−1} [4], [5], [6]. The 2011 Thailand megaflood, 2020 Jakarta monsoon floods and 2021 Kuala Lumpur flash floods are recent reminders that

the exposure is multi-hazard—coastal, fluvial and pluvial—and concurrent [7], [8].

These same cities anchor global trade. Maritime transport carries over 80% of world trade by volume, and ASEAN hosts both top-tier ports (Singapore, Port Klang, Tanjung Priok, Laem Chabang) and the Strait of Malacca chokepoint. Ports are critical infrastructure with a multi-year planning phase and a 50-year-plus useful life, so an asset designed today will operate well into the zone of large climate uncertainty [9], [10]. Present-day multi-hazard damage to global port infrastructure already has an expected value at risk of about \$7.6 billion per year (range \$4.0–17.4 billion), and roughly 80% of ports are exposed to flooding—coastal, rainfall and riverine [11]. First-order flood-screening is the cheapest input to the resulting trillion-dollar adaptation question, yet it is exactly what under-resourced ASEAN municipalities cannot obtain.

Existing tools fall into three categories, none of which simultaneously offers high resolution, multi-hazard scope, open code, open data and per-country calibration (Table I). The only 30 m three-hazard global model is commercial [12], and its free tranche excludes the major ASEAN megacities. Bespoke municipal studies are closed and not reproducible. Open global alternatives are an order of magnitude coarser ($\sim$ 10 km) and lack the pluvial layer that drives most urban flood damage in the region [13], [14], [15]. The result is an equity gap: open, high-resolution, multi-hazard flood information is unavailable to the agencies that most need it.

This paper addresses that gap. We present an open-source, open-data, per-country-calibrated 30 m multi-hazard flood pipeline and atlas for four ASEAN cities, validate that it *locates real floods*, and—following the revision of an earlier draft—tighten every claim to the evidence that supports it. We also reframe the contribution explicitly as forward-looking decision support: in a “new climate era,” planning by extrapolating historical flood frequencies systematically underprices tail risk, and a scenario- and return-period-conditioned screen is the corrective.

TABLE I
COMPARATOR LANDSCAPE FOR ASEAN-RELEVANT FLOOD-SCREENING TOOLS.

Tool	Res.	Hazards	Open code/data	Per-country IDF
This work	30 m	C + F + P	Yes / Yes	Yes (4 services)
Fathom 3.0 [12]	30 m	C + F + P	No / No	Global synth. [†]
Aqueduct 4.0 [13], [14]	~10 km	C + F	Method / Yes	None
GLOFRIS [15]	~10 km	F (+C)	Yes / Yes	None
City studies	High	Various	No / No	Per-city

[†]Fathom’s free tranche (accessed and verified 2025; pin a dated snapshot at review) is restricted to selected developing countries and excludes the ASEAN megacities covered here. C=coastal, F=fluvial, P=pluvial.

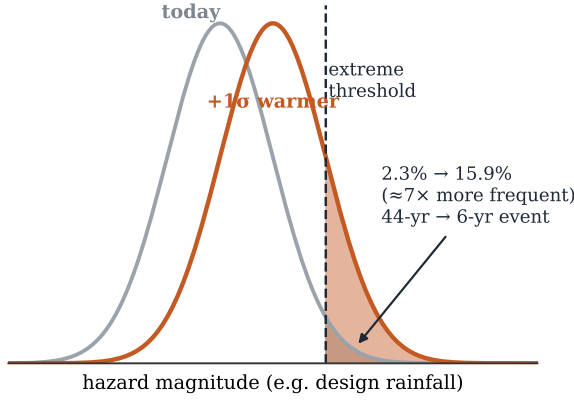


Fig. 1. Climate “math” for screening. A modest shift of the hazard distribution produces a large, non-linear increase in the frequency of threshold exceedances—here $\approx 7\times$. Static extrapolation of the historical record under-prices the tail; return-period, scenario-conditioned screening does not.

II. DECISION CONTEXT: WHY SCREENING MUST BE FORWARD-LOOKING

Infrastructure optimized for the twentieth-century climate now faces conditions outside its design envelope—roads buckling in unprecedented heat, drainage overwhelmed by rainfall beyond prior high-water marks [16]. The core reason is a property of shifting distributions, not of any single forecast. As a hazard distribution shifts warmer, the frequency of *extremes* grows much faster than the mean. Fig. 1 makes this concrete: a mean shift of one standard deviation raises the probability of exceeding a “ $+2\sigma$ ” threshold from 2.3% to 15.9%—roughly a $7\times$ jump that turns a 44-year event into a 6-year event, even though the average moved only modestly [17]. A planner who screens flood risk by extrapolating the historical record therefore systematically under-prices the tail. The corrective is to screen on return periods and explicit climate scenarios—which is exactly the surface our atlas provides.

The decision relevance is sharpened by the long life of the assets at risk (Fig. 2). A port entering planning today is completed in the mid-2030s and operates into the 2080s, by which point SSP-pathway divergence is wide: median ASEAN SLR by 2100 spans $+0.62$ to $+1.62$ m, and high-end ranges are

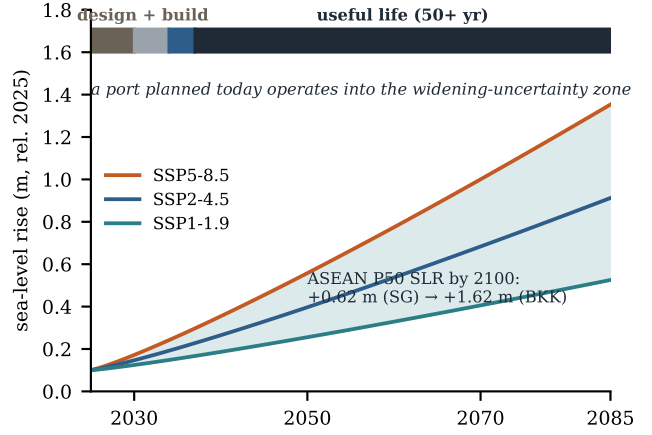


Fig. 2. Asset life versus scenario uncertainty. A port planned today operates through the 2080s, deep inside the widening SLR fan. The screen exists to make that fan—across SSP and horizon—legible to a planner before capital is committed.

larger still. Two further realities frame how the screen will be used. First, adaptation planning lags decarbonization: among the largest global ports, 89% (31/35) have mitigation plans but only 66% (23/35) have adaptation plans [9]. Second, hard remediation is slow and costly—the U.S. Coastal Texas (“Ike Dike”) barrier is projected at \$57 billion over 20 years [9]—so a cheap, open, comparable screen is the rational first move before any such commitment. Global port adaptation to SLR alone is estimated at \$223–768 billion by 2050; deciding *where* to spend it begins with screening.

III. OPEN MULTI-HAZARD PIPELINE

A per-city configuration drives a single orchestration script (Fig. 3). Every input is free and most require no registration: Copernicus GLO-30 DEM, ERA5-Land precipitation [18], UHSLC tide gauges, GloFAS v4 discharge reanalysis [19], IPCC AR6 SLR projections [2], ESA WorldCover 2021 [20], and OpenStreetMap waterways. The implementation is pure Python on the scientific stack; no licensed data or external binaries are required, which is what makes third-party rebuild from scratch possible.

Per-country IDF anchoring. The most important in-region calibration is the pluvial design rainfall. Global synthetic statistics under-represent tropical-convective extremes. We quantify this per service as the relative shortfall of the global-synthetic depth against the national IDF curve at matched duration and return period; across the four services (PUB, JPS-MSMA, TMD-RID, BMKG) the deficit spans 28–62%. As an illustrative single cell, a national 6-hour 100-year design depth of ~ 250 mm against a global-synthetic ~ 155 mm is a 38% shortfall—a gap that propagates directly into ponding depth. We therefore fit a two-anchor Gumbel distribution to each country’s published standard. We stress that two (return period, depth) anchors *exactly determine* Gumbel’s two parameters: this is exact determination, not a regression fit, so RP100 depends entirely on anchor placement. We anchor at each service’s published 2-year and (where available) 100-year

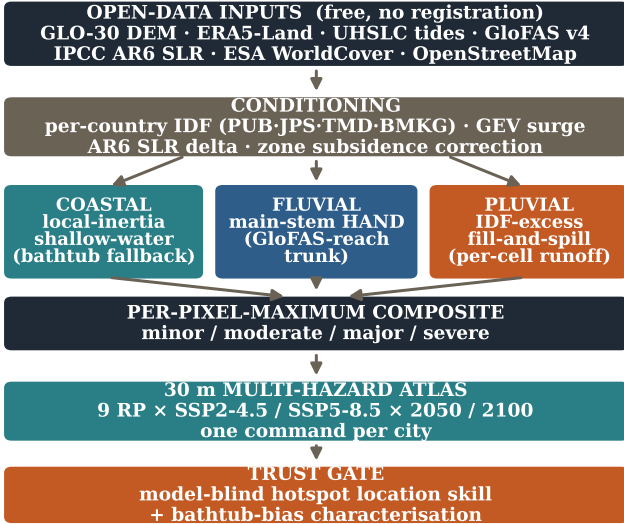


Fig. 3. Pipeline data flow: free open-data inputs → per-hazard solvers (inertial coastal, main-stem-HAND fluvial, fill-and-spill pluvial) → per-pixel-maximum composite and severity classification → trust gate.

design points, making RP100 a near-interpolation where a 100-year point is published and a mild extrapolation otherwise; the storm duration is matched to the dominant local mechanism (1-hour secondary-drain for Singapore’s sub-hourly convective bursts, 6-hour elsewhere).

Three hazards. *Coastal*: a GEV fit to UHSLC annual-maximum storm-surge residuals (T_TIDE-detided [21]), consistent with global surge reanalyses [22], [23], plus the AR6 SLR delta, datum-aligned via a CMEMS mean-dynamic-topography offset, and routed by a local-inertia shallow-water solver [24] on a staggered grid; a connectivity bathtub solver is retained only as a fallback where the sea is enclosed within the DEM domain. *Fluvial*: the GloFAS design discharge at the matched return period is converted to a design stage through a Manning-type uniform-flow rating on the reach-trunk geometry, with bankfull stage subtracted for rivers carrying permanent baseflow; the residual is mapped via Height Above Nearest Drainage (HAND) [25], [26] referenced to the main-stem trunk the modelled discharge represents (for Kuala Lumpur, the $\geq 180 \text{ km}^2$ accumulation trunk), not a channel-initiation threshold or raw OSM network, which over-broaden the floodplain. We flag a scale caveat: GloFAS at its native resolution is a coarse instrument for an urban trunk, so fluvial outputs are screening-grade. *Pluvial*: the IDF excess is routed by a catchment-routed fill-and-spill cascade [27] (depressions resolved by a Planchon–Darboux fill [28]) with a per-cell runoff coefficient from WorldCover. This is a steady-state method without hydrograph timing or infiltration dynamics; we report it as a known structural limit of the pluvial layer (Sec. V). The three depth rasters are composited by per-pixel maximum.

Scenarios and subsidence. All hazards are produced for SSP2-4.5 and SSP5-8.5 at 2050 and 2100 on a

TABLE II
COMBINED FLOOD EXTENT AT RP100, SSP5-8.5 / 2100 (km^2). COASTAL SHOWN AS BATHTUB → INERTIAL.

City	Coast.	Fluv.	Pluv.	Comb. [‡]	Dominant
Singapore	67 → 48 [§]	112*	19	184	canal + coast
Kuala Lumpur	0 → 0	126	268	362	pluvial
Bangkok	3,546 → 283	788	433	≈ 900	fluvial
Jakarta	159 → 112	389	169	560	fluv. + pluv.

Inertial coastal = bathtub × (inertial bias / bathtub bias) from Fig. 4.

[‡]Combined uses the *inertial* coastal layer; Bangkok combined is the headline correction (bathtub combined was 3,598). [§]Singapore’s documented present-day coastal baseline is near zero, so its residual ratio is an artifact (Sec. V). *Singapore’s “fluvial” layer is PUB primary canal-overflow under long-duration design rainfall, not natural-river flooding—a definitional caveat that limits cross-city comparability of this column.

subsidence-corrected DEM. The zone-based correction applies documented post-2013 rates—up to $\sim 25 \text{ cm yr}^{-1}$ in the worst north-Jakarta zones and $\sim 1\text{--}2 \text{ cm yr}^{-1}$ in Bangkok—accumulated to the horizon. In north Jakarta this contributes on the order of 1 m by 2050, rivaling or exceeding the SLR signal, so the subsidence component is reported separately and flagged for sensitivity rather than buried in the forcing.

IV. THE OPEN FLOOD ATLAS

At the canonical RP100/SSP5-8.5/2100 combination the atlas shows the expected physical contrast between cities (Table II): low-lying Bangkok is coastal-dominated under the raw solver, Jakarta fluvial/pluvial, and inland Kuala Lumpur pluvial. Crucially, the headline combined extent is sensitive to the coastal solver. The bathtub coastal layer over-predicts (Sec. V), so we present coastal extents *both* ways. Substituting the inertial layer collapses Bangkok’s combined extent from 3,598 to $\approx 900 \text{ km}^2$ —it ceases to be coastal-dominated and becomes fluvial-bounded. A planner should read the inertial column; the bathtub column is shown only to expose the correction.

The mitigation delta, corrected. The cleanest single number for an adaptation planner is the mitigation delta—the flooded area avoided by meeting the lower-emissions pathway. On the *bathtub* layer, the avoided Bangkok coastal RP100 land (SSP2-4.5 vs SSP5-8.5 at 2100) is -133 km^2 . We previously claimed such deltas are “robust to absolute bias.” That is true only for additive offsets. The bathtub bias is *multiplicative* ($\sim 12.5\times$ for Bangkok), so the difference of two inflated extents is itself inflated: computed on the corrected inertial layer the avoided area is $\approx 11 \text{ km}^2$, an order of magnitude smaller. The *sign* and policy direction (avoidance under lower emissions) are robust; the *magnitude* is not bias-free and must be read off the inertial layer. We report both.

V. VALIDATION AND TRUSTWORTHINESS

An open model is useful only if it can be trusted to flood where flooding actually occurs. We test this with a *pre-defined*, model-blind documented-hotspot location-skill gate. For each city we freeze, before consulting any model output,

TABLE III
DOCUMENTED-HOTSPOT GATE (PRESENT-DAY, RP100, ≥ 0.10 M, 50 M).
PASS = $HR \geq 0.70 \wedge CRR \geq 0.70 \wedge TSS\ CI > 0$.

City	pos/dry	HR	CRR	TSS [95% CI]	verdict
Kuala Lumpur	17/7	0.76	0.86	0.62 [0.25, 0.88]	PASS
Singapore	38/20	0.82	0.65	0.47 [0.21, 0.72]	sig.; $CRR < .70$
Bangkok	16/7	0.56	0.86	0.42 [0.04, 0.75]	sig.; $HR < .70$
Jakarta	18/8	0.89	0.50	0.39 [0.03, 0.75]	sig.; $CRR < .70$

Jakarta is reported *both ways*: before the model-blind reclassification of two record-documented controls (Menteng, Gambir; inundated 2007 and 2013) TSS was 0.16 (no skill); after, 0.39. The reclassification was fixed on flood-record evidence *before* the first gate run, never to make the gate pass.

a register of localities with documented flooding (positives) and localities documented to have stayed dry (controls), each geocoded and DEM-verified. The combined present-day wet mask (pluvial $\vee$ fluvial $\vee$ coastal, ≥ 0.10 m, within a 50 m hit radius matching geocoding precision) is scored against the register at the event-matched RP100, reporting hit-rate (HR; sensitivity), correct-reject-rate (CRR; specificity) and the true-skill statistic ($TSS = HR + CRR - 1$) with a BCa bootstrap 95% confidence interval (10,000 resamples); the 2×2 association is corroborated by Fisher’s exact test.

Two reported bars, stated explicitly. We distinguish two criteria that an earlier draft conflated. *Significant discriminative skill* is the weaker condition that the TSS confidence interval excludes zero. *PASS* is the stricter, *pre-registered* bar $HR \geq 0.70$ and $CRR \geq 0.70$ and $TSS\ CI > 0$. All four cities meet the weaker condition; one (Kuala Lumpur) meets PASS (Table III). Because the gate is run in four cities, we note the family-wise context: Kuala Lumpur’s PASS survives a Bonferroni-adjusted $\alpha = 0.0125$, and so does the pooled register. We also acknowledge the small registers (24/58/23/26 points): lower CI bounds of 0.03–0.04 for Bangkok and Jakarta are statistically significant but fragile, and we present them as such rather than as robust.

Order of operations (no retuning). Two design choices materially affect the gate, so we state their timing explicitly. (i) The main-stem-HAND trunk threshold (≥ 180 km² for Kuala Lumpur) was fixed by the physics of which channel the modelled discharge represents; it corrects a false positive on a 60–77 m hill and yields a credible floodplain. (ii) The Jakarta control reclassification was decided on the 2007/2013 flood record. Both were fixed *before* the first gate evaluation and anchored to documented facts, not gate output; a genuine control that the model floods stays in the register as a reported false positive. We therefore report Jakarta’s TSS both unreclassified (0.16) and reclassified (0.39) and carry both forward, rather than presenting only the favorable number.

Bathtub bias is corrected, not merely cheap. A bathtub solver—the default in open screening tools—over-predicts documented present-day coastal inundation by 1.7–25 $\times$ at RP100, because 30 m terrain cannot resolve the sub-pixel road raises, canals and bunds that protect small present-day events. Replacing it with the local-inertia solver, where the sea

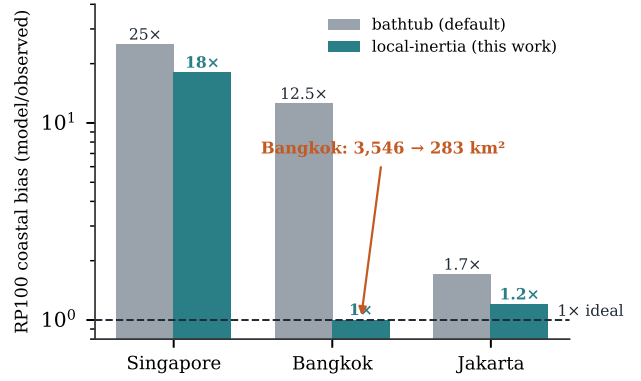


Fig. 4. Bathtub-bias factor (model/observed) at RP100 by city, with the local-inertia correction. The 12.5 $\times$ Bangkok reduction to $\approx 1 \times$ (3,546 $\rightarrow$ 283 km²) is the headline; Singapore’s residual ratio stays high only because its documented present-day coastal extent is near zero.

connects to the domain boundary, brings the over-prediction to $\approx 1 \times$ (Fig. 4): Bangkok’s RP100 coastal extent drops from 3,546 to 283 km² (a 12.5 $\times$ reduction). We benchmark this corrected layer against *coastal-specific* references (storm-surge and high-tide inundation footprints), not the 2011 compound megaflood, whose fluvial headwaters lie outside the model domain; the 2011 extent is reported only as a loose upper-envelope sanity check. The fix is solver-architecture, not data: Singapore’s residual 18 $\times$ ratio reflects a near-zero documented baseline, not a model failure.

VI. FROM SCREENING TO STRATEGIC DECISIONS

The atlas is a decision instrument, and its value tracks the decisions it informs. Three uses follow directly. *Triage*: 80% of ports face flooding exposure and present-day port damage risk is $\sim \$7.6$ B yr⁻¹, with 32% concentrated in cyclone-prone clusters such as the U.S. Gulf and Southeast Asia [11]; a comparable cross-city surface lets an agency rank its own sites before commissioning costly bespoke studies. *Scenario framing*: because the same solver and inputs are used for every city and scenario, cross-scenario deltas (read off the corrected layer) isolate the adaptation-relevant signal from absolute bias. *Sequencing under uncertainty*: with hard barriers costing tens of billions and taking decades—and with high-tide flooding days at exposed ports projected to rise 12–25 $\times$ by 2050 [9]—an open screen supports the choice between building now for a projected future and preserving option value to retrofit later. This is the practical content of “climate intuition”: quantifying how the hazard distribution and the human response interact, and planning for the tail rather than the mean [16].

The equity argument is sharpened by this framing. The cities our atlas covers are not marginal: they host the region’s largest ports and the Malacca chokepoint, exactly the assets a commercial model’s free tranche omits. Lowering first-order flood information from a procurement question to a software install is therefore not only a humanitarian good but a trade-resilience one.

VII. IMPACT AND LIMITATIONS

The intended use is humanitarian: open, validated, reproducible hazard maps as decision support for municipal disaster-management, adaptation and port-planning agencies that cannot license commercial models. The model is honest as a screening *upper bound* under three explicit assumptions: no active pumping, no sub-pixel defences resolved by the 30m DEM, and per-pixel-maximum (marginal, not joint-exceedance) composition. Two findings bound transferability and are reported rather than tuned away: single-stage HAND does not transfer to flat deltas fed by an out-of-domain mega-river (Bangkok, Jakarta) [29], which bounds Bangkok’s HR; and the pluvial fill-and-spill solver is steady-state and currently heterogeneous across cities (a documented homogenization step), which explains residual specificity loss from over-ponding on elevated ground. Manila (enclosed bay) and Ho Chi Minh City (enclosed delta) are deferred because their enclosed-sea topology breaks the inertial solver’s wall condition; relaxing that condition is the principal prerequisite for extending the bias-aware coastal treatment.

VIII. CONCLUSION

An open-source, open-data, per-country-calibrated 30m multi-hazard flood atlas for ASEAN cities is feasible, reproducible from free data, and—by a pre-defined model-blind location-skill gate—demonstrably skilful in the four cities tested, with its coastal over-prediction structurally corrected and every headline number now read off the corrected layer. By targeting exactly the cities that commercial high-resolution models exclude, and by framing the output as forward-looking decision support for critical infrastructure under deep climate uncertainty, the release is offered as a public good for humanitarian flood-risk reduction and an invitation to extend.

REPRODUCIBILITY

Code, per-city configuration and the frozen validation registers (with their pre-reclassification labels) are released openly through an anonymized mirror for double-blind review; the de-anonymized repository is provided on acceptance, enabling third-party rebuild of every result from free data.

REFERENCES

- [1] Asian Development Bank, “Southeast Asia and the economics of global climate stabilization,” 2022.
- [2] B. Fox-Kemper et al., “Ocean, Cryosphere and Sea Level Change,” in *IPCC AR6 WGI*, Cambridge Univ. Press, 2021.

- [3] G. Lenderink et al., “Scaling of extreme rainfall with temperature,” 2017.
- [4] D. Chaussard et al., “Sinking cities in Indonesia: ALOS PALSAR detects rapid subsidence,” *Remote Sens. Environ.*, vol. 128, pp. 150–161, 2013.
- [5] H. Z. Abidin et al., “Land subsidence of Jakarta and its relation with urban development,” *Nat. Hazards*, vol. 59, pp. 1753–1771, 2011.
- [6] N. Phien-wej et al., “Land subsidence in Bangkok, Thailand,” *Eng. Geol.*, vol. 82, pp. 187–201, 2006.
- [7] S. Hallegatte et al., “Future flood losses in major coastal cities,” *Nat. Clim. Change*, vol. 3, pp. 802–806, 2013.
- [8] B. Tellman et al., “Satellite imaging reveals increased proportion of population exposed to floods,” *Nature*, vol. 596, pp. 80–86, 2021.
- [9] J.P. Morgan, “Sea Change: Port infrastructure, climate risks and the future of global trade,” Climate Advisory, 2025.
- [10] S. E. Hanson and R. J. Nicholls, “Demand for ports to 2050: climate policy, growing trade and the impacts of sea level rise,” *Earth’s Future*, vol. 8, no. 8, e2020EF001543, 2020.
- [11] J. Verschuur, E. E. Koks, S. Li et al., “Multi-hazard risk to global port infrastructure and resulting trade and logistics losses,” *Commun. Earth Environ.*, vol. 4, no. 5, 2023.
- [12] O. E. J. Wing et al., “A 30m global flood inundation model for all climates,” 2024 (free-access tier accessed 2025; pin dated snapshot).
- [13] H. Hofste et al., “Aqueduct Floods methodology,” WRI Tech. Note, 2019.
- [14] P. J. Ward et al., “Aqueduct Floods: global flood risk maps and analysis,” 2020.
- [15] E. H. Sutanudjaja et al., “PCR-GLOBWB 2,” *Geosci. Model Dev.*, vol. 11, pp. 2429–2453, 2018.
- [16] J.P. Morgan, “Navigating the New Climate Era: Building Intuition for Strategic Decision-Making,” Climate Intuition, 2025.
- [17] R. N. Patel, D. B. Bonan and T. Schneider, “Changes in the frequency of observed temperature extremes largely driven by a distribution shift,” *Geophys. Res. Lett.*, vol. 51, no. 24, e2024GL110707, 2024.
- [18] J. Muñoz-Sabater et al., “ERA5-Land,” *Earth Syst. Sci. Data*, vol. 13, pp. 4349–4383, 2021.
- [19] L. Alfieri et al., “A global network for operational flood risk reduction (GloFAS),” *Environ. Sci. Policy*, vol. 84, pp. 149–158, 2018.
- [20] D. Zanaga et al., “ESA WorldCover 10m 2021 v200,” Zenodo, 2022.
- [21] R. Pawlowicz, B. Beardsley and S. Lentz, “Classical tidal harmonic analysis with error estimates in MATLAB using T_TIDE,” *Comput. Geosci.*, vol. 28, pp. 929–937, 2002.
- [22] S. Muis et al., “A global reanalysis of storm surges and extreme sea levels,” *Nat. Commun.*, vol. 7, 11969, 2016.
- [23] S. Muis et al., “A high-resolution global dataset of extreme sea levels, tides and surges,” *Front. Mar. Sci.*, vol. 7, 263, 2020.
- [24] P. D. Bates, M. S. Horritt and T. J. Fewtrell, “A simple inertial formulation of the shallow water equations,” *J. Hydrol.*, vol. 387, pp. 33–45, 2010.
- [25] A. D. Nobre et al., “Height Above the Nearest Drainage,” *J. Hydrol.*, vol. 404, pp. 13–29, 2011.
- [26] B. Schwanghart and D. Scherler, “TopoToolbox 2,” *Earth Surf. Dynam.*, vol. 2, pp. 1–7, 2014.
- [27] R. Barnes, K. L. Callaghan and A. D. Wickert, “Computing water flow through complex landscapes – Part 2: Fill–Spill–Merge,” *Earth Surf. Dynam.*, vol. 8, pp. 431–445, 2020.
- [28] O. Planchon and F. Darboux, “A fast, simple and versatile algorithm to fill depressions of DEMs,” *Catena*, vol. 46, pp. 159–176, 2002.
- [29] N. Trinh et al., “Flood risk in the Mekong and Chao Phraya deltas,” 2017.